# Javis Tze Yie Kho

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## Profile

- BSc Physics graduate at Durham University with an interest in Medical Physics and Condensed Matter Physics.
- Studied Atomic and Optical Physics, Condensed Matter Physics (with Soft Matter Physics), Nuclear Physics and Statistical Physics.
- Offer holder of MSc Medical Physics at the University of Aberdeen

## Languages

- English - C1
- Chinese - C2
- Malay - B1
- Korean - A1

## Education

- **Durham University** 2023 – 2026
  BSc Physics: Upper Second Class (2:1)
- **Lodge International School** 2021 – 2022
  A-Level: Physics (A), Chemistry (A), Mathematics (A), Further Mathematics (B)
- **SMK Kuching High** 2016 – 2021
  Malaysian Certificate of Education: 5A+, 3A and 1A-

## Skills

### Lab Skills and Electronics

- Capable of expanding on concepts introduced in class through independent study and applying advanced knowledge to conduct experiments collaboratively with other physicists.
- Practical experience in a wide range of physical fields, including classical mechanics (kinematics and fluid mechanics), optics, astronomy and condensed matter physics.
- Experienced and confident in operating electronics such as oscilloscopes and lock-in amplifiers.

### Programming and Digital Skills

- Skilled in using Python and Microsoft Excel for data analysis, calculations and creating professional-quality graphs for scientific reports.
- Proficient in using LaTeX to write, organise and format scientific reports and other documents such as CVs.
- Proficient in using photo and video editing software such as Adobe Lightroom and DaVinci Resolve.

### Time Management Skills and Multi-tasking

- Able to balance academics study with co-curricular activities such as cinematography projects and archery.
- Adopted to high pressure environment, prioritising tasks to maintain productivity and work-life balance.
- Delivered high-quality lab reports under tight deadlines while concurrently managing two major physics research projects, preparing workshop materials in advance, completing assignments ahead of schedule and attending regular section heads' meetings at PalTV.

### Communication and Teamworks

- Delivered practical training sessions to cinematography beginners, clearly explaining how to operate broadcast equipment including gimbals, cameras, image transmitters/receivers and microphones.
- Delegated tasks as the Deputy Head of Broadcasting at PalTV and Captain of the DU Archery Team, coordinating team members and maintaining clear communication to achieve shared objectives.

- Experienced in collaborative group work, contributing constructively in multidisciplinary and diverse team environments.

## Projects

**1. Thesis - Rydberg Physics In Quantum Microwave-Optical Converter: A Quantitative Analysis**

- Reviewed, compared and analysed the development of microwave-optical converters across multiple platforms, including electro-mechanical-optic systems, magnons, Rydberg atoms and excitons.
- Developed advanced skills in literature review, technical paper writing and project time-management through independent thesis research.

**2. Research-Led Investigation in Astronomy**

- Quantified the effects of solar perturbation, mutual interactions between the Galilean moons, and Jupiter's oblateness on the inferred mass of Jupiter using a simplified dynamical model.
- Operated astronomical telescopes and processed observational data, including archival datasets, using DS9 and GAIA in Linux operating system.

**3. Advanced Laboratory in Condensed Matter Physics**

- Determined the band-gap energies of zinc selenide and silicon using visible-light spectroscopy and investigated their temperature dependence through experimental data analysis and model fitting in Python.
- Demonstrated independent learning and the ability to actively recall and apply concepts in atomic physics, condensed matter physics and optics.
- Demonstrated leadership by delegating tasks and explaining key concepts to lab partner, ensuring effective teamwork and clear communication.

## Extra-curriculum

**1. PalTV**

- Deputy Head of the Broadcasting and camera/drone operator in this student-led media society.
- Collaborated with the Cinematography and Engineering Departments to produce high-quality broadcasts for events such as DCI and DUCE.
- Operated cameras and drones during regional tournaments, including Regatta 24, Regatta 25 and Barbarian versus Team DU, demonstrating advanced cinematography skills.

**2. Archery**

- Captain and Co-President of the Durham University Archery Club (DUAC).
- Managed sponsorships, promoted the club during Freshers'/Orientation Week and represented Durham University at the regional BUCS 2024 archery tournament.
- Trained beginners in archery techniques, fostering skill development and teamwork.

**3. First Aider at St. John Ambulance England**

- Community First Aider for the North East region.
- Acquired practical first aid skills, including emergency response, patient assessment and basic life support.